

VEDANT PAWAR

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PROFESSIONAL SUMMARY

Results-driven Python Developer and AI/ML Engineer with hands-on experience building production-grade NLP systems, LLM-integrated backends, and real-time Computer Vision applications (YOLO, OpenCV) using TensorFlow, PyTorch, and Keras. Architected the backend for a 7-domain autonomous AI robot using FastAPI, LangChain, RAG pipelines, and Docker, driving 25% workflow efficiency gains through optimized object detection and image processing. Proficient in cloud deployment, REST APIs, and scalable system design, with experience in front-end development, video analysis, and dataset creation, actively seeking impactful Python/AI backend or ML Engineer roles.

CORE TECHNICAL SKILLS

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| • Languages & Backends: Python | • FastAPI | • Flask |
| • Django | • JavaScript | • REST APIs |
| • SQL (MySQL) | • PostgreSQL | • Async Programming. AI / ML / NLP: LLM Integration |
| • LangChain | • RAG Pipelines | • Prompt Engineering |
| • Vector Databases (ChromaDB) | • Hugging Face | • OpenCV |
| • YOLO | • SSD | • Faster R-CNN |
| • NLP | • Sentiment Analysis | • TensorFlow |
| • PyTorch | • Keras. Voice & UI: gTTS | • ElevenLabs API |
| • Gradio | • Streamlit | • React (Basics) |
| • HTML5 | • CSS3. DevOps & Cloud: Docker | • Docker Compose |
| • AWS (EC2) | • S3 deployment patterns) | • Git |
| • GitHub | • CI/CD Pipelines (GitHub Actions) | • Cloud Platforms. Core CS Concepts: Data Structures & Algorithms (DSA) |
| • OOP | • Microservices | • Scalable Architecture |
| • API Design Patterns | • Backend Optimization | |
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PROFESSIONAL EXPERIENCE

AI/ML & IoT Intern – Incubators (CREVA AI Robot Project) Jan 2026 – Mar 2026

- **Pipeline Architecture:** Designed NLP-based interaction pipelines for the CREVA AI Robot across 7 functional domains, enabling real-time AI request processing and improving overall workflow efficiency by 25% (reducing average task completion time from ~40s to ~30s) utilizing Computer Vision techniques with OpenCV and PIL/Pillow, and leveraging object detection algorithms such as YOLO, SSD, and Faster R-CNN.
- **API Optimization:** Integrated 5+ AI modules with web interfaces via RESTful APIs, reducing inter-service API latency by ~30% through payload optimization and connection pooling.
- **Scalable Deployment:** Designed and implemented a scalable AI+IoT backend architecture supporting modular, hot-swappable deployment workflows, enabling 3 new AI modules to be added without system downtime, utilizing Docker containerization and cloud platforms.

- Containerization: Containerized the FastAPI backend using Docker, standardizing local development and enabling consistent, reproducible deployment environments across a 5-person engineering team.
 - Agile Execution: Collaborated in an Agile/Scrum environment across 6 two-week sprints, maintaining 100% on-time sprint delivery while iterating on real-time AI system integration.
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KEY PROJECTS

CREVA AI System – Multi-Domain Autonomous AI Robot

- Multi-Domain Orchestration: Deployed NLP-driven interaction pipelines enabling real-time, context-aware AI communication across 7 independent functional domains, serving 20+ concurrent request types and incorporating Computer Vision with YOLO, SSD, and Faster R-CNN, and utilizing video analysis and annotation tools for dataset creation.
 - LLM Integration: Integrated multiple LLM and AI APIs into a unified, scalable FastAPI backend with LangChain orchestration, supporting hot-swappable modular AI workflows and reducing integration effort by ~40%.
 - System Reliability: Engineered modular backend architecture with clean separation of concerns containerized with Docker, implementing robust API communication with error handling, retry logic, and async request queuing to achieve 99%+ uptime during active live testing.
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EDUCATION

Bachelor of Computer Applications (BCA) 2022 – 2025

Shri Shivaji Science College, Amravati, Maharashtra

- Relevant Coursework: Data Structures, Algorithms, Database Systems, Computer Networks, and programming fundamentals, providing a solid foundation for front-end development, video processing, and image analysis.
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CERTIFICATIONS & TRAINING

• Python for Everybody – Coursera

- FastAPI & REST API Development – Self-directed (3 production projects completed)
- LangChain for LLM Application Development – DeepLearning.AI (In Progress)
- AWS Cloud Practitioner Essentials – (In Progress; EC2 & S3 deployment patterns applied in projects)
- Docker & Containerization – Applied across 3 production GitHub repositories, with experience in cloud platforms, video analysis, and dataset creation.